SegmentIQ AI

Customer Intelligence & Behavioral Segmentation

DASHBOARD PRINT VIEW

All dashboard sections included — Executive Overview, Customer Explorer, Behavioral Segmentation, Segment Profiles, Customer Value & RFM, Marketing Strategy, Campaign Simulator, Data Management, and Export & Reports.

This is a visual, print-friendly rendering of the supplied dashboard's interface and feature set. Interactive controls are represented as static printed content; the original HTML remains the interactive version.

Data note: The source prototype generates synthetic sample data by default. The dashboard’s displayed figures depend on the live browser calculations, filters, selected cluster count, RFM thresholds, and any uploaded dataset. This PDF does not claim to be a screenshot or a live interactive export.

Contents

#	Dashboard section	What is shown
01	Executive Overview	KPI cards, segment charts, segment comparison
02	Customer Explorer	Searchable customer table, profile fields, recent transactions
03	Behavioral Segmentation	K-means feature set, centroid comparison, cluster sizes
04	Segment Profiles	Behavior evidence, hypotheses, opportunities, channels, risks
05	Customer Value & RFM	Editable-threshold concept, RFM group metrics
06	Marketing Strategy	Segment-level objectives, campaigns, channels, success metrics
07	Campaign Simulator	Scenario comparison, editable assumptions, formulas
08	Data Management	CSV input schema, validation rules, data export
09	Export & Reports	Available report and data export options

01 · Executive Overview

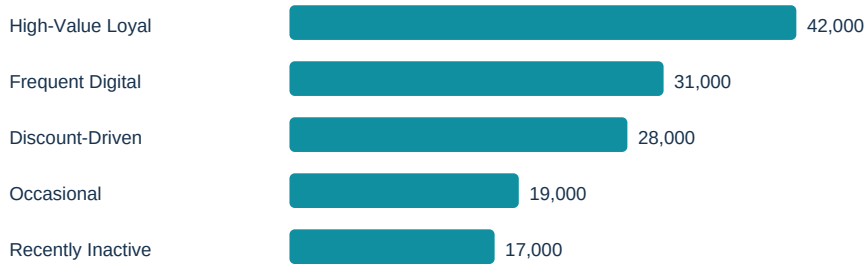
SegmentIQ AI · Understand Behavior. Discover Segments. Personalize Growth.

The overview uses KPI cards and visual comparisons to summarize customer value and behavior. Values below are illustrative display examples, not calculated live output from your browser session.

1,000	5	$—	—%	20%	—%
Total customers*	Example segments*	Avg. customer value	Repeat purchase rate	High-value definition	At-risk / lost share

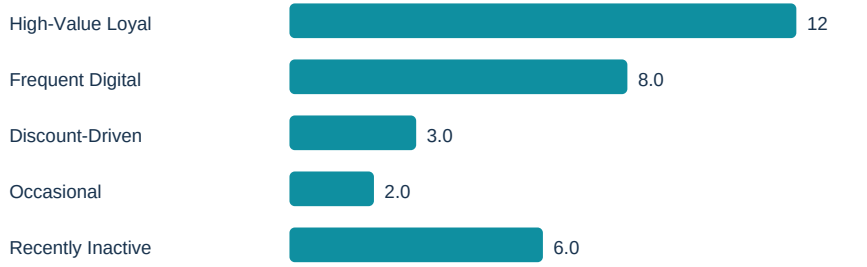
Customers by segment

Revenue by segment



Average order value

Purchase frequency (orders/customer)



02 · Customer Explorer

SegmentIQ AI · Understand Behavior. Discover Segments. Personalize Growth.

Search controls: customer ID / segment search · sortable customer, segment, RFM group, spend, AOV, orders, recency, discount, engagement. The browser dashboard shows up to 200 matches at a time.

Customer records — table layout

Customer	Behavioral segment	RFM group	Spend	AOV	Orders	Recency d	Discount	Engagement	Recommended action
C0001	High-Value Loyal Customers	Champions	$1,840	$153.33	12	18	8%	82	Offer early access / VIP perk
C0002	Frequent Digital Shoppers	Loyal / Active	$735	$91.88	8	27	12%	76	Send app recommendations
C0003	Discount-Driven Buyers	Potential / Low-Engagement	$285	$95.00	3	74	67%	38	Test capped offer with holdout
C0004	Occasional Customers	Potential / Low-Engagement	$160	$80.00	2	112	0%	24	Send category-based reminder
C0005	Recently Inactive Customers	At-Risk	$510	$85.00	6	205	10%	18	Win-back message, then suppress
C0006	Mixed-Behavior Customers	Potential / Low-Engagement	$390	$97.50	4	63	25%	49	Include in A/B tests

Selected customer detail panel

Profile fields: customer ID, assigned segment, RFM group, region, acquisition channel, age band.

Purchase metrics: transaction count, total spend, average order value, days since last purchase, discount rate, return rate.

Interaction summaries: web visits, email engagement, app use, support contacts, and loyalty membership when available.

Recent transactions: up to eight latest rows with date, category, amount, discounted flag, and returned flag.

The rows above are example layout data for demonstrating the dashboard view; they are not records extracted from a real customer database.

03 · Behavioral Segmentation

SegmentIQ AI · Understand Behavior. Discover Segments. Personalize Growth.

Methodology and controls

Algorithm: K-means clustering with k-means++ initialization, fixed seed 42, and up to 80 iterations. Cluster count k is configurable from 2 to 8.

Preprocessing: features are z-scored; frequency and spend are log-transformed. Missing engagement is mean-imputed for clustering, with a warning. IDs, demographics, signup date, and acquisition channel are excluded.

Cluster names: assigned by rules applied to computed centroid values. The label rules are heuristic; they do not validate a customer's motivation.

Features used by the clustering model

Feature	Meaning
Recency	Days since most recent transaction
Frequency (log)	Log-transformed order count
Spend (log)	Log-transformed total spend
Average order value	Spend divided by order count
Discount rate	Share of transactions with discount applied
Return rate	Share of transactions marked returned
Engagement	Composite score from web, email, and app summaries
Category diversity	Number of distinct categories purchased

Example centroid comparison (z-scores)

Segment	Recency	Frequency	Spend	AOV	Discount	Return	Engagement	Diversity
High-Value Loyal	+0.1	+1.3	+1.5	+0.8	-0.4	-0.2	+0.6	+0.5
Frequent Digital	-0.2	+0.7	+0.4	+0.2	-0.1	+0.0	+1.4	+0.6
Discount-Driven	+0.0	-0.1	-0.2	-0.1	+1.6	+0.2	+0.1	+0.0
Occasional	+0.2	-1.1	-0.7	-0.1	+0.0	-0.1	-0.5	-0.4
Recently Inactive	+1.3	+0.3	+0.2	+0.0	-0.2	+0.0	-0.8	+0.2

Illustrative centroid values shown for visual explanation only. The actual dashboard computes centroids from the currently loaded dataset. The clusters are fitted on the full loaded dataset; filters affect display, not the fit.

04 · Segment Profiles

SegmentIQ AI · Understand Behavior. Discover Segments. Personalize Growth.

Segment	Objective	Channels	Contact cadence	Success metric	Risks
High-Value Loyal Customers	Retain and grow share of wallet	Email, app push	2–3 contacts/month	Repeat rate; spend/customer	Reward cost; over-contact
Frequent Digital Shoppers	Increase app-led repeat purchase	App, web, email	Weekly	App conversion; orders/active user	Notification fatigue
Discount-Driven Buyers	Protect margin while retaining volume	Email, SMS	1–2/month, event-led	Incremental gross profit/customer	Margin erosion; needs holdout
Occasional Customers	Raise purchase frequency	Email	1–2/month	Second-purchase rate; 90-day frequency	Limited purchase history
Recently Inactive Customers	Win back lapsed customers	Email, SMS	2–3 touches then suppress	Reactivation; net contribution	Reason for lapse unknown
Mixed-Behavior Customers	Learn what drives value	Email	About 1/month	Response rate by test cell	Segment may not be distinctive

What each profile displays

- Customer share, revenue share, AOV, average orders, recency, discount rate, engagement, and repeat rate.
- Top product categories and deviations from the overall average.
- A clearly labeled unvalidated hypothesis about possible behavior.
- Marketing opportunity, channel, contact frequency, message themes, and risks.

Segment names are assigned from centroid rules. The profile hypothesis is not validated evidence about individual intent.

05 · Customer Value & RFM Analysis

SegmentIQ AI · Understand Behavior. Discover Segments. Personalize Growth.

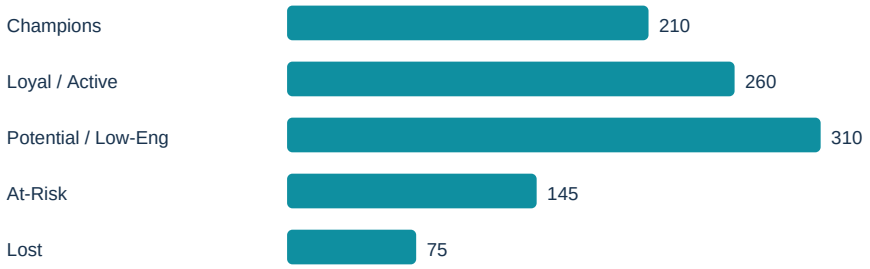
RFM is a separate rule-based system from the K-means behavioral clusters. Recency, Frequency, and Monetary scores use configurable thresholds from 1 to 4.

30 / 90 / 180
Recency cut-offs (days)

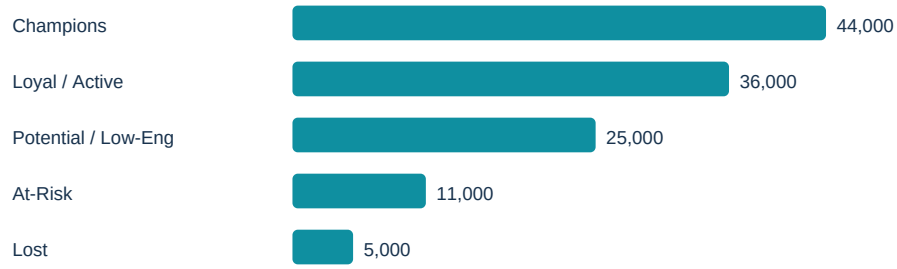
2 / 5 / 10
Frequency cut-offs (orders)

$200 / $600 / $1,500
Monetary cut-offs (spend)

Customers by RFM group



Revenue by RFM group



RFM group	Rule summary
Champions	Combined RFM score ≥ 10
Loyal / Active	Recency score ≥ 3 and frequency score ≥ 2, if not already Champions
At-Risk	Recency score ≤ 2 and frequency score ≥ 3, if not classified earlier
Lost	Recency score = 1, if not classified earlier
Potential / Low-Engagement	Fallback group

Counts, revenue, and repeat rates in the live dashboard are derived from the currently filtered data and current threshold settings. The charts here illustrate the dashboard presentation, not live-calculated figures.

06 · Marketing Strategy

SegmentIQ AI · Understand Behavior. Discover Segments. Personalize Growth.

The strategy module translates segment evidence into testable campaign recommendations. It shows priority, evidence, objective, campaign, channel, success metric, and limitations.

Segment	Priority	Campaign direction	Channels	Success metric	Limitations
High-Value Loyal	High	Loyalty rewards, early access, complementary cross-sell, VIP recognition	Email, app push	Repeat rate; spend/customer	May subsidize purchases that would happen anyway
Frequent Digital	Medium	App recommendations, personalized browsing, relevant digital campaigns	App, web, email	App conversion; orders per active user	Engagement may not convert to spend
Discount-Driven	Medium	Targeted promotion tests; measure incremental profit, not just volume	Email, SMS	Incremental gross profit/customer	Holdout needed; margin erosion risk
Occasional	Medium	Personalized recommendations, category emails, purchase reminders	Email	Second-purchase rate; 90-day frequency	Low history limits personalization
Recently Inactive	High	Tailored win-back reminders; suppress after repeated non-response	Email, SMS	Reactivation rate; net contribution	Lapse reason unknown; incentive may be unnecessary
Mixed-Behavior	Low	Controlled tests across offers and channels	Email	Response rate by test cell	Segment may not be distinctive

Ask the data · local rule-based Q&A;

Supported question areas: revenue contribution, discount sensitivity, engagement, high-value marketing, re-engagement, and product categories.

How it works: keyword rules read the currently filtered data and return predefined analyses. It is not a connected LLM.

Known gap: an engagement decline over time cannot be measured directly without interaction time-series data.

07 · Campaign Simulator

SegmentIQ AI · Understand Behavior. Discover Segments. Personalize Growth.

The simulator compares No Campaign, Standard, and Targeted Personalized scenarios. Inputs are editable assumptions; results are not validated predictions or causal estimates.

Editable assumptions

Input	Meaning	Personalization effect
Budget ($)	Maximum campaign budget	Same budget constraint
Customers targeted	Eligible target size	Limited by budget / cost
Response rate (%)	Share of targeted customers who respond	Multiplied by response uplift
Conversion rate (%)	Share of responders who convert	Applied to responses
Average order value ($)	Expected order value	Used to calculate gross revenue
Gross margin (%)	Gross margin assumption	Applied to revenue
Cost per customer ($)	Campaign cost per targeted customer	Multiplied by personalization cost multiplier
Response uplift (×)	Assumed personalization lift	Applied to response rate
Cost multiplier (×)	Relative personalized contact cost	Applied to per-customer cost

Scenario comparison — illustrative output layout

Scenario	Targeted	Responses	Conversions	Revenue	Gross profit	Cost	Net contribution	ROI
No Campaign	—	—	—	$0	$0	$0	$0	n/a
Standard	500	25	5	$500	$175	$1,000	-$825	-82.5%
Targeted Personalized	500	33	7	$700	$245	$1,200	-$955	-79.6%

Formula reference

Targeted = min(customers, budget ÷ cost per customer) · Responses = targeted × response rate · Conversions = responses × conversion rate · Revenue = conversions × AOV · Gross profit = revenue × gross margin · Net contribution = gross profit – campaign cost · ROI = net contribution ÷ campaign cost.

Revenue is gross revenue, not verified incremental revenue. Test with a randomized holdout group and measure incremental profit after costs before scaling.

08 · Data Management

SegmentIQ AI · Understand Behavior. Discover Segments. Personalize Growth.

Default sample data

The source HTML creates a deterministic illustrative dataset using seed 20261028, with 1,000 synthetic customers and generated transactions, interaction summaries, categories, regions, channels, age bands, discounts, and returns. The date-generation logic begins from November 1, 2025.

Do not present synthetic sample data as real customer evidence. Upload a valid transaction CSV to replace the sample for analysis.

CSV field	Requirement	Behavior / notes
customer_id	Required	Customer identifier; aliases such as customerid, cust_id, customer recognized.
transaction_date	Required	Date; aliases such as date and order_date recognized.
purchase_amount	Required	Must be numeric and greater than zero; aliases amount, total, order_value recognized.
transaction_id	Optional	Duplicate transaction IDs are excluded; order_id and txn_id aliases recognized.
product_category	Optional	Defaults to Uncategorized when absent.
discount_applied	Optional	Recognizes 1/true/yes/y as true.
returned	Optional	Recognizes 1/true/yes/y as true.
location_region	Optional	Defaults to Unknown; region alias recognized.
acquisition_channel	Optional	Defaults to Unknown; channel alias recognized.
age_band	Optional	Defaults to Unknown.

Validation and export actions

- Exclude rows with empty customer ID, non-positive/non-numeric amount, invalid date, or duplicate transaction ID.
- Preview source rows and show valid, invalid, excluded, and duplicate counts with exclusion reasons.
- Download loaded transactions as CSV; restore original synthetic sample; replace the current dataset with valid imported records.
- Export customer data with segment, RFM group, orders, spend, AOV, recency, discount, return, engagement, region, and channel.

Uploaded transaction-only data does not include web/email/app interaction summaries unless those fields are separately supplied; these fields may show as n/a.

09 · Export & Reports

SegmentIQ AI · Understand Behavior. Discover Segments. Personalize Growth.

Export option	Contents
Full Dashboard PDF	Executive summary, KPIs, revenue contribution, segmentation methodology, segment comparison and profiles, RFM, marketing recommendations, campaign scenario, limitations and next steps.
Executive Summary PDF	Headline metrics and revenue contribution by segment.
Segment Profiles PDF	Segment evidence, top categories, behavioral deviations, and unvalidated hypotheses.
Marketing Recommendations PDF	Segment priorities, objectives, campaign directions, channels, success metrics, and limitations.
Customer Data CSV	Customer ID, segment, RFM group, orders, total spend, AOV, recency, discount rate, return rate, engagement, region, and channel.
Transactions CSV	Transaction ID, customer ID, date, product category, amount, discount flag, and returned flag.
Print-Friendly Report	Uses browser print behavior for a print view.

Limitations shown in the dashboard

- Synthetic default data; no real-world conclusions should be drawn from sample figures.
- Clusters depend on k, selected features, preprocessing, and dataset; segment labels are heuristic.
- Mean imputation for missing engagement can reduce information quality.
- Segment motivations and hypotheses are not validated by transaction data alone.
- Campaign simulator is assumption-driven and not a causal prediction engine.
- Local Q&A; uses keyword rules and supports a narrow set of topics; it is not an LLM.
- RFM groups depend on editable thresholds and are separate from K-means clusters.

Recommended next steps

Connect approved CRM and interaction data; document KPI definitions; audit missingness and duplicates; test multiple cluster counts and stability; validate campaigns using holdout experiments; monitor drift, opt-outs, complaints, data quality, and privacy.

This document is a print-friendly visual rendering based on the supplied segmentiq-ai.html. Interactive controls are static in PDF; use the original HTML file to interact with filters, customer selection, CSV upload, simulator inputs, and native export buttons.